

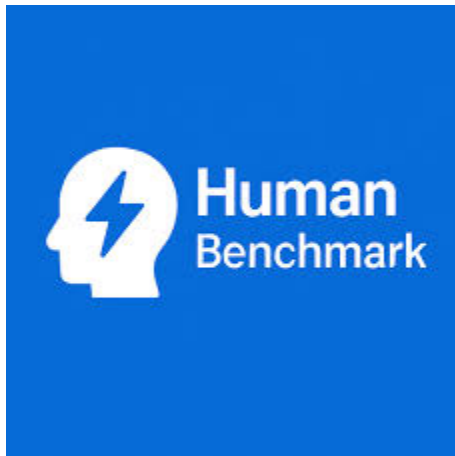
Element B V1

# MyDesign Portfolio

## *Element B Initial Template - AT - 6 points*

### Element B: Documentation and Analysis of Current and Prior Design Attempts

#### Current or Prior Design Attempt #1 - <Human Benchmark>

|         |                                                                                                                                                                                                                                                                                           |
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| Source  | Mobile reaction time applications such as "Reaction Time Test" and "Human Benchmark-style apps" are widely documented in digital health and UX testing discussions, including reviews of mobile sensor limitations (Mobile Tech Review, 2022).                                            |
| Image   |  <p><a href="https://play.google.com/store/apps/details?id=xyz.mmcode.humanbenchmark.human_benchmark">https://play.google.com/store/apps/details?id=xyz.mmcode.humanbenchmark.human_benchmark</a></p> |
| Details | Smartphone reaction time apps measure how quickly a user responds to a visual or                                                                                                                                                                                                          |

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|                 | auditory stimulus using a touchscreen interface. A stimulus (e.g., color change or sound) appears after a random delay, and the user taps the screen as quickly as possible. The app calculates reaction time based on system timestamps and touch input detection (Mobile Tech Review, 2022). Some versions include multiple trials, averages, and difficulty adjustments.                                                                                                                                          |
| <i>Analysis</i> | Although highly accessible, smartphone-based systems suffer from hardware latency, including touchscreen sampling delay and operating system processing lag. These introduce systematic error, reducing measurement precision. External factors such as grip style, screen protector thickness, and background processes also influence response time variability. As a result, while suitable for informal tracking, these systems lack the calibration consistency required for controlled scientific measurement. |

## Prior Design Attempt #2 - <Ruler drop test>

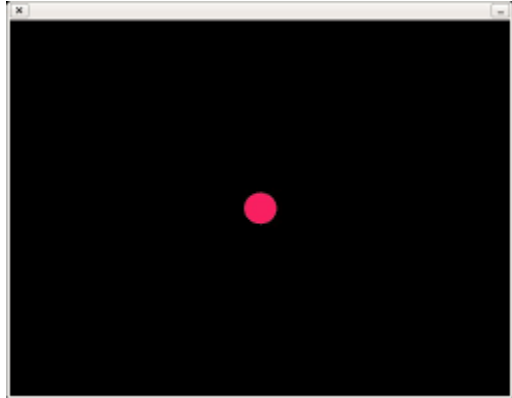
|               |                                                                                                                                                                                            |
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| <i>Source</i> | This method is standard in educational physics experiments and is documented in university-level laboratory manuals and science outreach organizations (Exploratorium, 2021; University of |
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|                 | Queensland Physics Lab Guide, 2020).                                                                                                                                                                                                                                                                                                                                                                                          |
| <i>Image</i>    |  <p><a href="https://www.scienceworld.ca/resource/reaction-time-ruler/">https://www.scienceworld.ca/resource/reaction-time-ruler/</a></p>                                                                                                                                                                                                   |
| <i>Details</i>  | The ruler drop test estimates reaction time by measuring the distance a ruler falls before being caught. This distance is converted into time using kinematic equations under gravitational acceleration assumptions. It is commonly used in classrooms to demonstrate human response latency and basic motion physics.                                                                                                       |
| <i>Analysis</i> | The ruler drop test is limited by high procedural variability, particularly in unpredictable release timing, which is manually controlled by the tester. It also measures a composite of visual processing and motor execution rather than isolating neural reaction time. Resolution is low because small differences in catch distance translate into relatively large time uncertainty, reducing experimental reliability. |

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### Prior Design Attempt #3 - Psychomotor Vigilance Task (PVT)

|          |                                                                                                                                                                                                                                                                                                                                         |
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| Source   | The PVT is a validated cognitive performance measure used in neuroscience and fatigue research, including NASA and sleep research institutions (Basner & Dinges, 2011; NASA Human Performance Research Program).                                                                                                                        |
| Image    |  <p><a href="https://en.wikipedia.org/wiki/Psychomotor_vigilance_task">https://en.wikipedia.org/wiki/Psychomotor_vigilance_task</a></p>                                                                                                              |
| Details  | The Psychomotor Vigilance Task is a computer-based test where participants respond to a stimulus appearing at random intervals. Reaction time is recorded across many trials to assess attention stability and alertness. It is widely used in aviation and medical fatigue monitoring due to its high sensitivity to cognitive lapses. |
| Analysis | The PVT is highly reliable and scientifically validated, but it is designed for sustained attention measurement rather than isolated reflex response. Its long duration                                                                                                                                                                 |

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|  | introduces fatigue effects that can skew later responses. Additionally, it requires controlled digital environments and standardized software, limiting portability and ease of deployment in non-laboratory settings. |
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### Prior Design Attempt #4 - <FitLight / BlazePod Reactive Training Systems >

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| Source | Commercial reactive training systems are used in sports science and athletic conditioning and are documented by manufacturers and sports performance research communities (FitLight Training Systems, 2023; BlazePod, 2024).                                                                                                                                                                                                                                          |
| Image  |  <p><a href="https://www.blazepod.com/products/mini-kit?currency=USD&amp;country=US&amp;variant=44166872170541&amp;stkn=1ac571fc37e7&amp;utm_source=Google&amp;utm_medium=cpc&amp;campaign">https://www.blazepod.com/products/mini-kit?currency=USD&amp;country=US&amp;variant=44166872170541&amp;stkn=1ac571fc37e7&amp;utm_source=Google&amp;utm_medium=cpc&amp;campaign</a></p> |

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|                 | <a href="#">_id=12789109824&amp;utm_content=193703657923&amp;utm_term=&amp;gad_source=1&amp;gad_campaignid=12789109824&amp;gbraid=0AAAAAC1jwrYg-4ovDJDVXViWW-rLnLxkx&amp;gclid=CjwKC Ajwt7XQBhBkEiwAtStpp0briCaPOiuhftqBEY-J8bKmsDR-ZS6lHUhMGAmdK4inORLaCl34hoCFEAQAvD_BwE</a>                                                                                                        |
| <i>Details</i>  | <p>These systems use wireless LED pods that illuminate in random or programmed sequences. Users must physically react by touching or moving to the illuminated target. Reaction time is recorded digitally via connected applications. They are widely used in elite sports training to improve reaction speed, agility, and cognitive-motor coordination.</p>                        |
| <i>Analysis</i> | <p>Although more advanced than app-based systems, these designs introduce confounding variables such as movement distance and athletic speed, meaning results are not purely reaction time. They also require multiple hardware units, making them expensive and less accessible. Setup complexity and spatial requirements further reduce usability in constrained environments.</p> |



## Element B V1

### Prior Design Attempt #5 - <Laboratory Response Box Systems (Cedrus / CANTAB) >

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| Source  | Professional response box systems are standard tools in cognitive neuroscience research, including Cedrus hardware and Cambridge Cognition CANTAB systems (Cedrus Corporation Technical Documentation, 2021; Cambridge Cognition, 2022).                                                                            |
| Image   |  <p><a href="https://www.crs ltd.com/tools-for-vision-science/subject-feedback/rb-540-series/nest/key-features">https://www.crs ltd.com/tools-for-vision-science/subject-feedback/rb-540-series/nest/key-features</a></p>        |
| Details | These systems use high-precision button response boxes connected to experimental software. Participants respond to controlled stimuli (visual or auditory), and reaction times are recorded with millisecond-level accuracy. They are widely used in clinical psychology, neuroscience, and human factors research. |

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| Analysis | These systems provide extremely high measurement accuracy and experimental control, making them a gold standard in research settings. However, they are expensive, non-portable, and require specialized software and controlled environments. Their complexity makes them unsuitable for everyday or educational use, limiting scalability. |
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## Rubric

| Element B: Documentation and Analysis of Prior Design Attempts                                                                 |                                                                      |                                                                   |                                                                                      |                                                                                          |                                                                              |                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
|                                                                                                                                | score = 5                                                            | score = 4                                                         | score = 3                                                                            | score = 2                                                                                | score = 1                                                                    | score = 0                                                                                   |
| <b>B1. Documentation of plausible prior attempts to solve the problem and/or related problems is drawn from _____.</b>         | a wide array of clearly identified and consistently-credible sources | a variety of clearly identified and consistently-credible sources | several—but not necessarily varied—clearly identified and generally-credible sources | a limited number of sources, some of which may not be clearly identified and/or credible | only one or two sources that may not be clearly identified and/or credible   | missing or minimal (a single source that is not clearly identified and/or credible)         |
|                                                                                                                                | score = 5                                                            | score = 4                                                         | score = 3                                                                            | score = 2                                                                                | score = 1                                                                    | score = 0                                                                                   |
| <b>B2. The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — _____.</b> | is consistently clear, detailed, and supported by relevant data      | is clear, generally detailed, and supported by relevant data      | is generally clear and contains some detail and relevant supporting data             | is overly general and contains little detail and/or relevant supporting data             | is vague and is missing any relevant details and/or relevant supporting data | cannot be inferred from information intended as analysis of past and/or current attempts to |



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|  |  |  |  |  |  | solve the<br>problem |
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